

✓ Zomato Data Analysis Project

```
!pip install matplotlib
```

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Requirement already satisfied: six>=1.5 in c:\users\puret\anaconda3\lib\site-packages (from py
```

```
!pip install seaborn
```

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Requirement already satisfied: six>=1.5 in c:\users\puret\anaconda3\lib\site-packages (from py
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Create the Dataframe

```
df=pd.read_csv("zomato_data.csv")
```

```
df.head()
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1/5	775	800	Buffet
1	Spice Elephant	Yes	No	4.1/5	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8/5	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300	Buffet

```
df.columns
```

```
Index(['name', 'online_order', 'book_table', 'rate', 'votes',
      'approx_cost(for two people)', 'listed_in(type)'],
      dtype='object')
```

```
df.columns.tolist()
```

```
['name',
 'online_order',
```

```
'book_table',
'rate',
'votes',
'approx_cost(for two people)',
'listed_in(type)']
```

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   name                                148 non-null    object
1   online_order                        148 non-null    object
2   book_table                          148 non-null    object
3   rate                                148 non-null    object
4   votes                               148 non-null    int64
5   approx_cost(for two people)         148 non-null    int64
6   listed_in(type)                     148 non-null    object
dtypes: int64(2), object(5)
memory usage: 8.2+ KB
```

df.head()

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1/5	775	800	Buffet
1	Spice Elephant	Yes	No	4.1/5	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8/5	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300	Buffet

✓ Convert the data type of column-rate

```
def handleRate(value):
    value=str(value).split('/')
    value=value[0]
    return float(value)

df['rate'] = df['rate'].apply(handleRate)
print(df.head())
```

	name	online_order	book_table	rate	votes	\
0	Jalsa	Yes	Yes	4.1	775	
1	Spice Elephant	Yes	No	4.1	787	
2	San Churro Cafe	Yes	No	3.8	918	
3	Addhuri Udupi Bhojana	No	No	3.7	88	
4	Grand Village	No	No	3.8	166	

	approx_cost(for two people)	listed_in(type)
0	800	Buffet
1	800	Buffet
2	800	Buffet
3	300	Buffet
4	600	Buffet

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   name                                148 non-null    object
1   online_order                        148 non-null    object
2   book_table                          148 non-null    object
3   rate                                148 non-null    float64
4   votes                               148 non-null    int64
5   approx_cost(for two people)         148 non-null    int64
6   listed_in(type)                     148 non-null    object
```

```
dtypes: float64(1), int64(2), object(4)
memory usage: 8.2+ KB
```

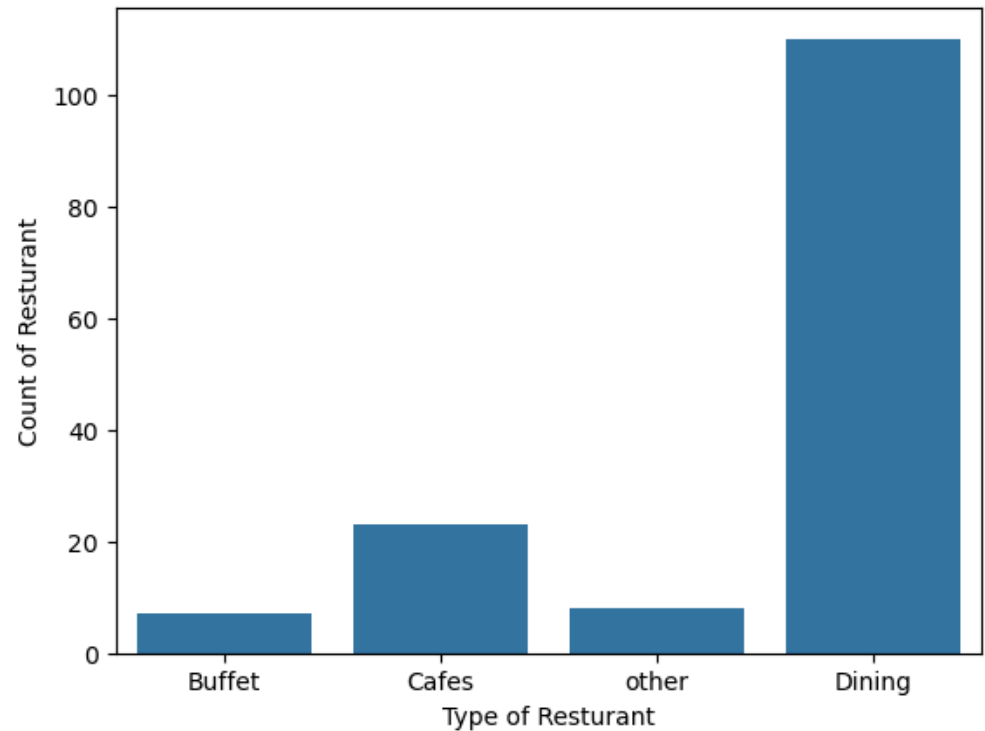
```
df.head()
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet

1.What type of Resturant do the majority of Customers order from?

```
sns.countplot(x =df['listed_in(type)'])
plt.xlabel("Type of Resturant")
plt.ylabel("Count of Resturant")
```

```
Text(0, 0.5, 'Count of Resturant')
```



Conclusion:Majority of the Resturant fals in Dining type

2.How many votes has each type of resturant recieved from customers?

```
group_data=df.groupby('listed_in(type)')['votes'].sum()
result=pd.DataFrame({'votes':group_data})
```

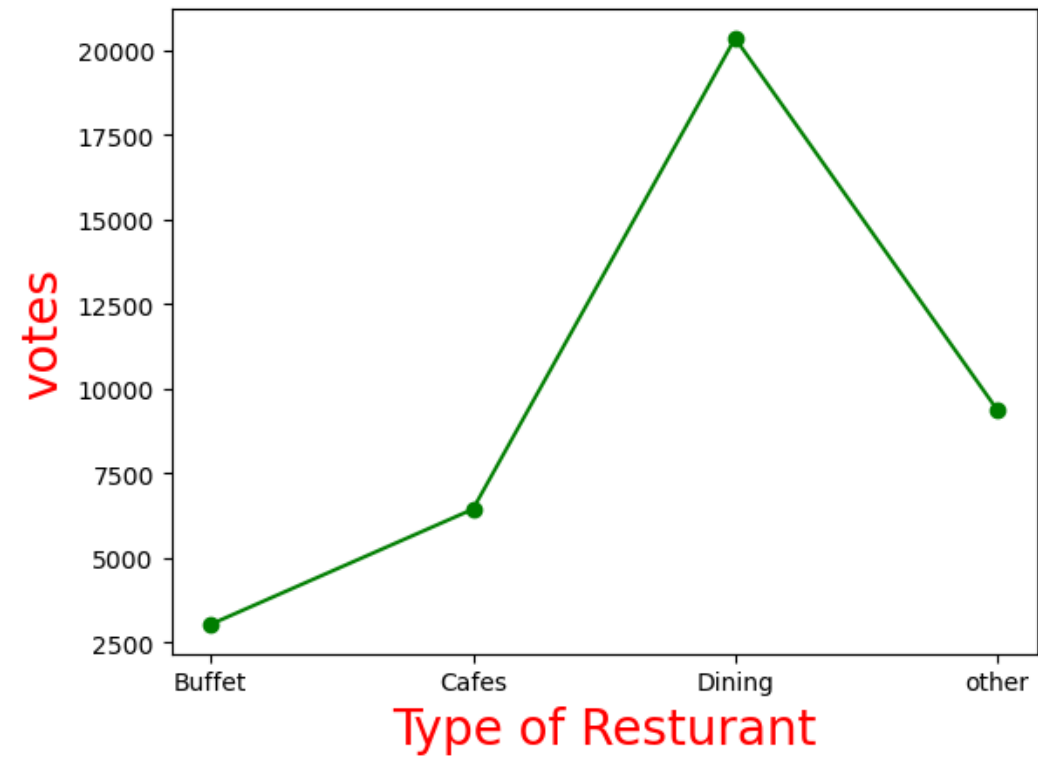
```
result
```

	votes
listed_in(type)	
Buffet	3028
Cafes	6434
Dining	20363
other	9367

```
plt.plot(result,c='green',marker='o')
plt.xlabel("Type of Resturant",c='red',size=20)
```

```
plt.ylabel("votes",c ='red',size=20)
```

```
Text(0, 0.5, 'votes')
```



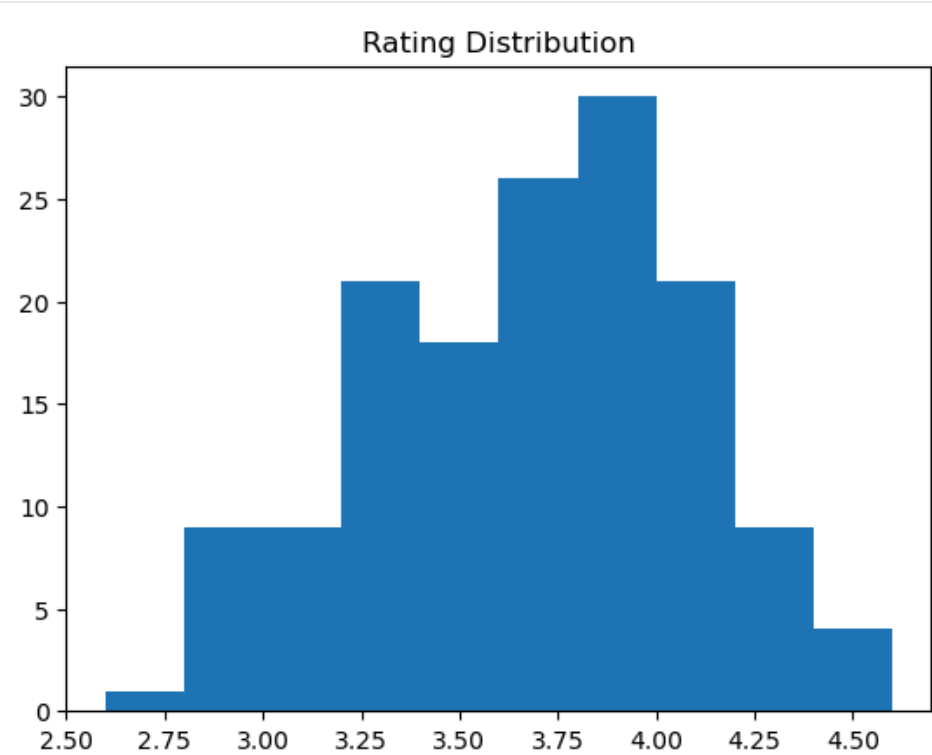
Conclusion:Dining resturants has recieved maximum votes

3.What are the ratings that the Majority of Resturants have Recieved?

```
df.head()
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
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```
plt.hist(df['rate'] ,bins=10)
plt.title("Rating Distribution")
plt.show()
```

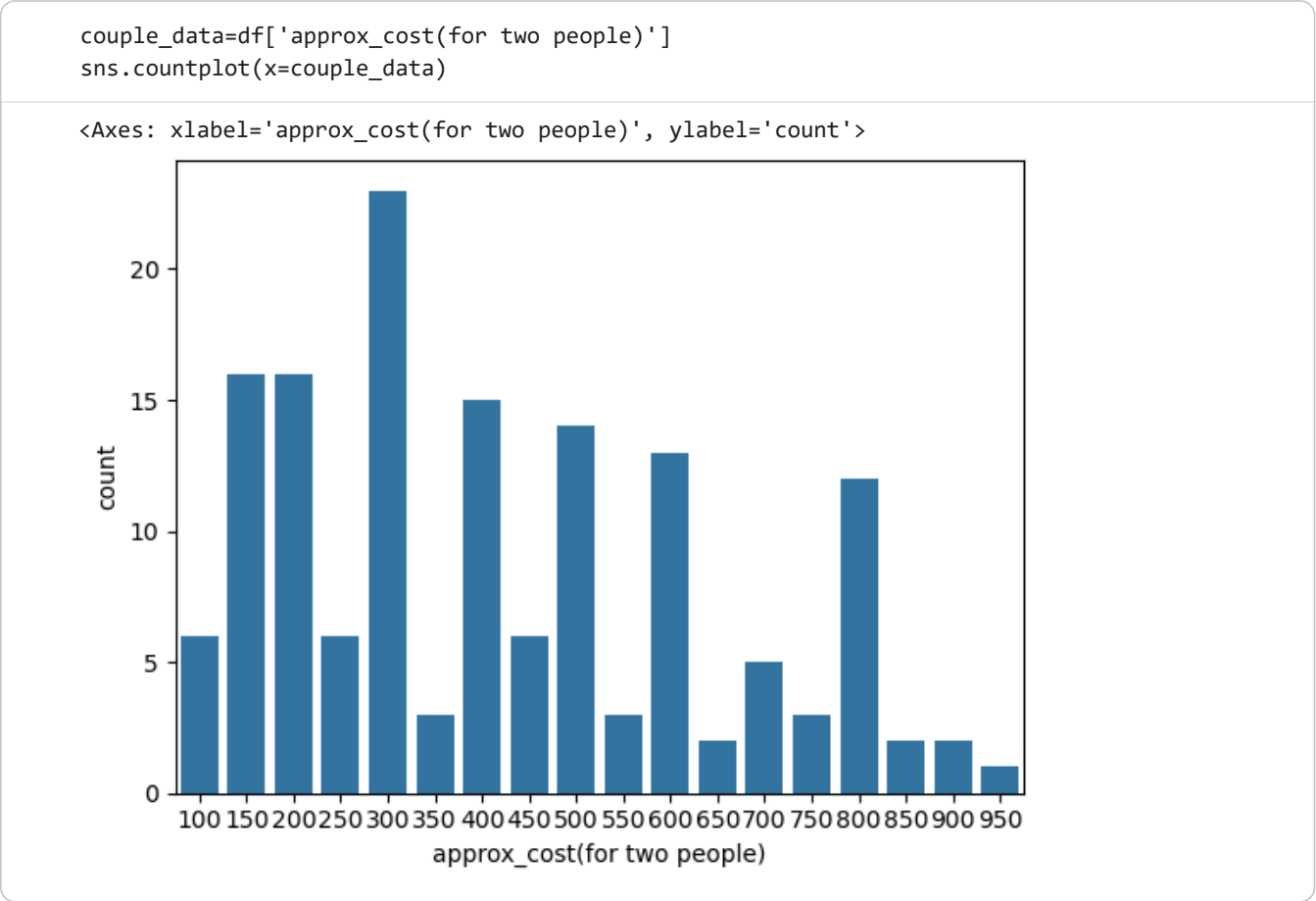


Conclusion:The Majority resturants recieved ratings from 3.75 to 4

4.Average order Spending by Couples?

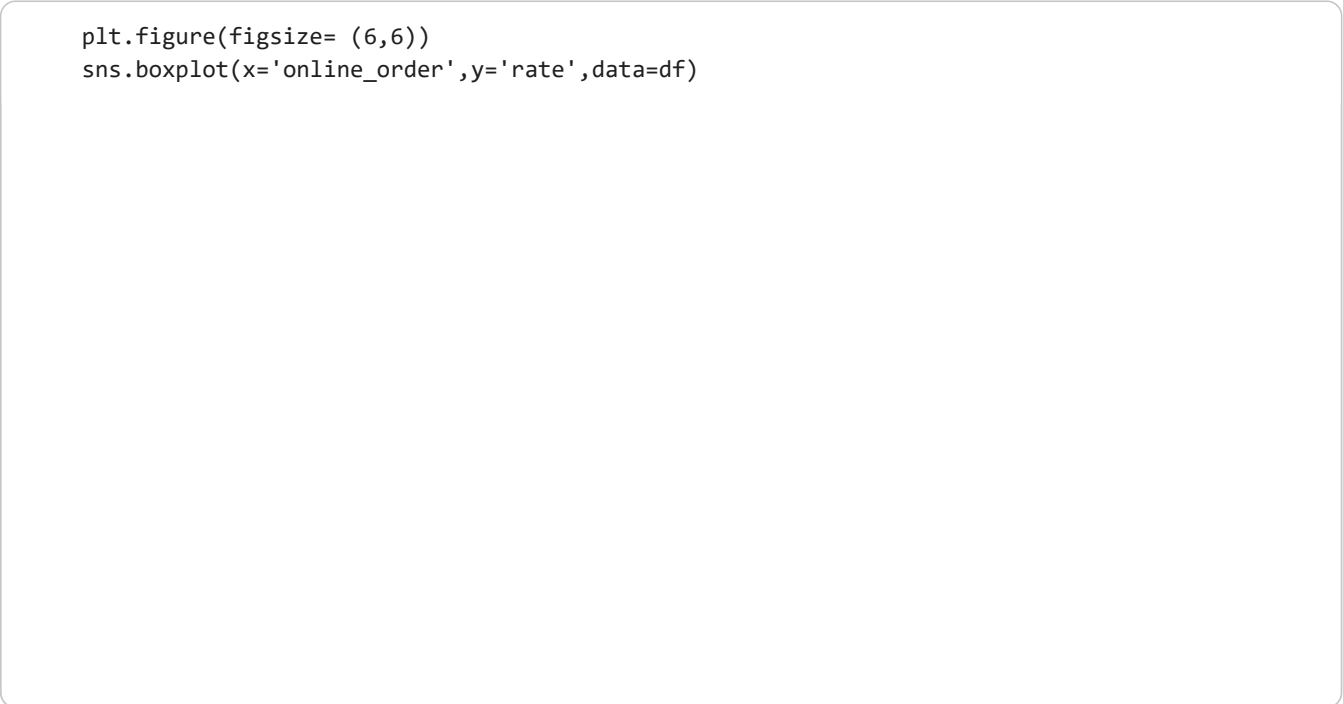
df.head()

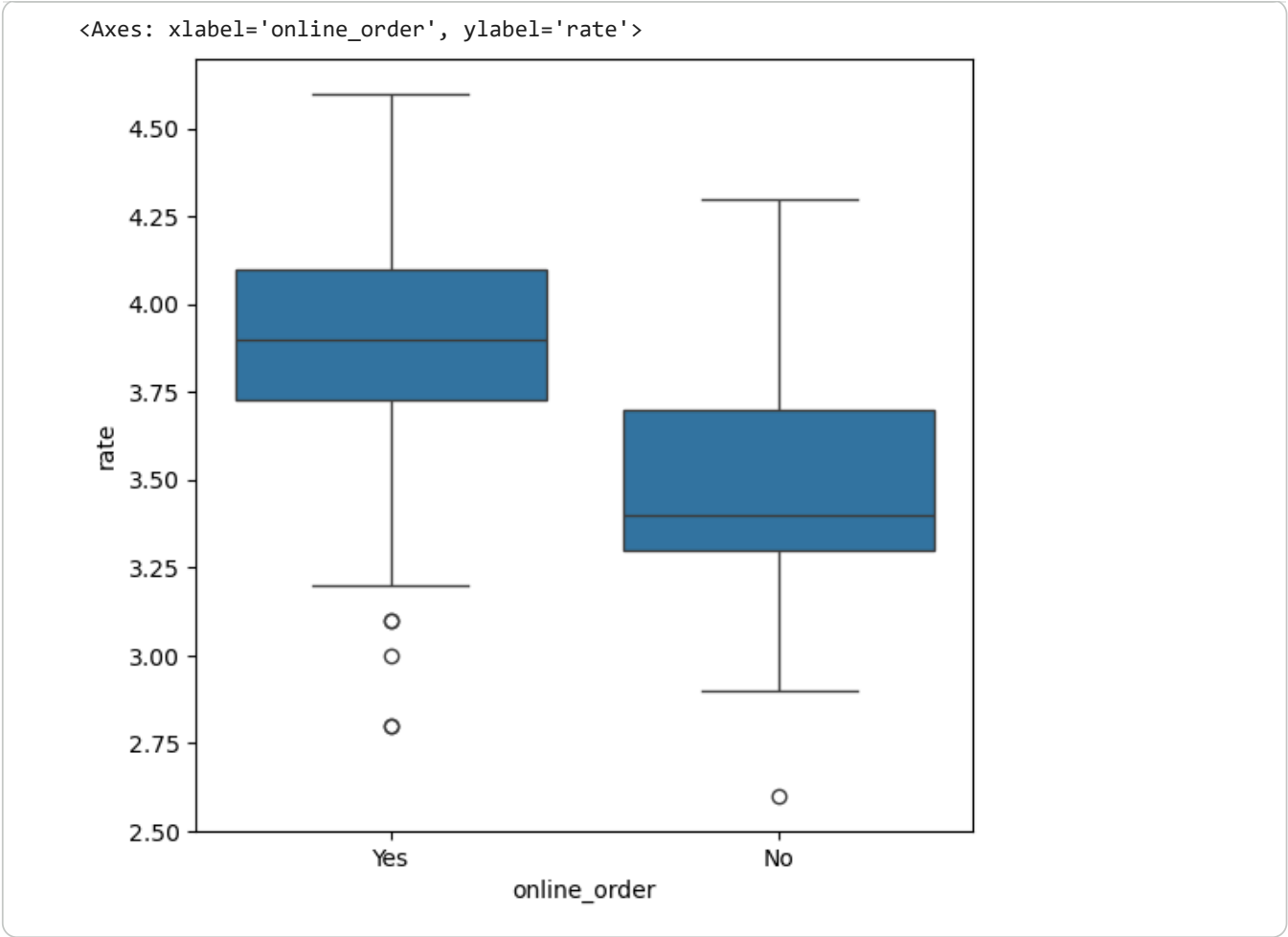
	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet



Conclusion:The Majority of couples prefers resturant with an approximate 300 rupees

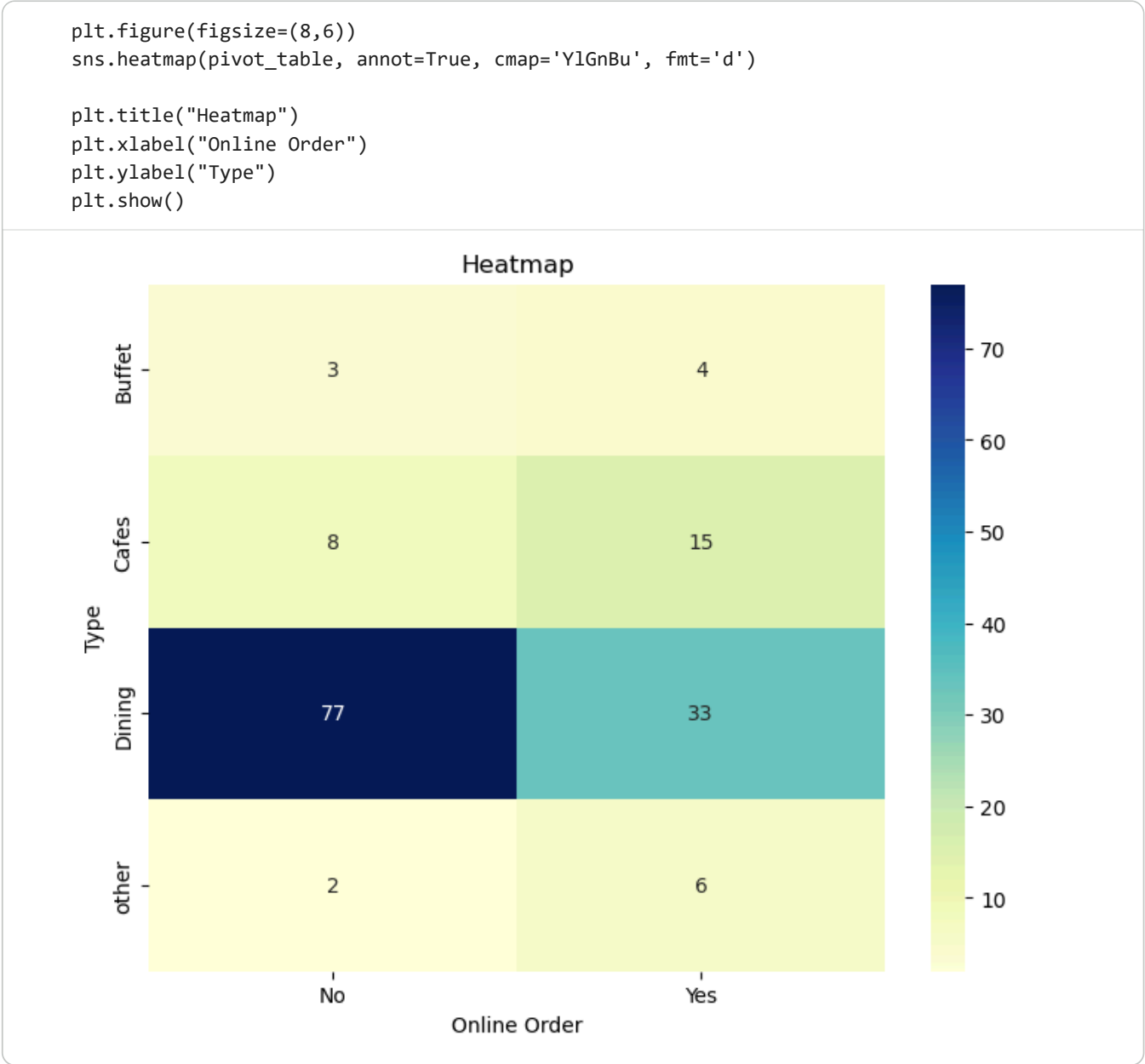
5.Which mode recieved maximum rating?





Conclusion:offline order recieved lower rating in comparsion to online orders

6.Which type of resturant recieved more offline orders,so that Zomato can provide Customers with some Good Offers?



pivot_table

online_order	No	Yes
listed_in(type)		
Buffet	3	4
Cafes	8	15
Dining	77	33
other	2	6

Start coding or [generate](#) with AI.

Start coding or [generate](#) with AI.